



Detoxification response in honey bee larvae exposed to agricultural intensification

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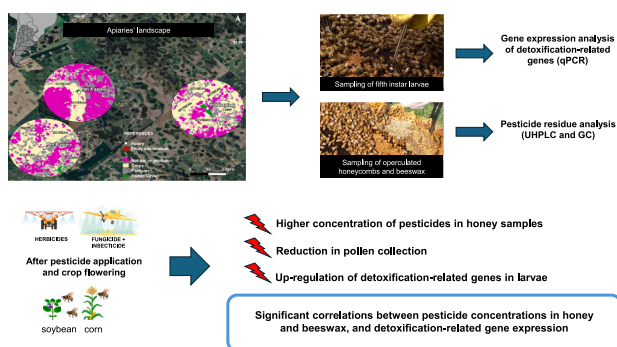
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HIGHLIGHTS

- The honey bee *Apis mellifera* is one of the main pollinators in agroecosystems.
- In these environments pollinators are exposed to combinations of agrochemicals.
- Expression of detoxification genes in larvae increases after pesticide application.
- Residues of agrochemicals in beeswax and honey samples were found.
- Pesticide concentrations correlated with the expression of two *CYP450* studied.

GRAPHICAL ABSTRACT



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ABSTRACT

Honey bee *Apis mellifera* colonies located in agroecosystems are exposed to pesticides and more fragmented habitats. The resources that bees obtain in these environments may be exposed to agrochemicals, which can accumulate in their colonies and be distributed among their nest mates. Hives placed in an agricultural setting located in the region of the Argentine Pampas were studied. Changes in the expression levels of insect cytochrome P450s, enzymes involved in the detoxification of xenobiotics, and the presence of pesticides in hive products at different times of crop management were evaluated. Our results showed that *CYP6AS2* and *CYP6AS4* expression in honey bee larvae increased significantly after crop flowering and pesticide application. Furthermore, residues of the herbicides atrazine and glyphosate, and the insecticide chlorantraniliprole were found in beeswax and honey samples collected from the same beehives, and their concentrations correlated with the

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expression profiles of *CYP6AS2*, *CYP6AS3* and *CYP9BD1*. These results underscore the potential risks of pesticides exposure to larval development, highlighting the need to mitigate agrochemical use in agricultural landscapes to safeguard honey bee colonies.

1. Introduction

In recent decades, planting of pollination-dependent crops has incremented, driving a growing demand for pollinators to sustain agricultural productivity (Aizen et al., 2008). However, global dependence on pollinators is increasing, and many agricultural practices continue to threaten bees and their essential pollination services (Aizen et al., 2019). Two of the major consequences of agricultural intensification across global landscapes are increased reliance on pesticides and declining biodiversity of both floral and pollinators (Decourtye et al., 2010). Bees can be exposed to pesticides by multiple routes. Oral exposure occurs by foraging on contaminated resources or by consuming contaminated food within the hive as immature or young adult bees (Farina et al., 2019; Naiara Gomes et al., 2020; Tomé et al., 2020). In addition, exposure by body contact can occur outside the hive, when bees forage for resources in the field during pesticide application, or inside the hive when they develop in contaminated wax cells or receive a diet with traces of pesticides (Johnson et al., 2009; Traynor et al., 2016). Although most studies focus on adult bees, the exposure of immature bees is equally critical and often overlooked, despite its potential to cause long-term effects (Macri et al., 2021; Tomé et al., 2020; Vázquez et al., 2023).

Grains, cereals, oil seeds, fruits, and vegetables rely on fungicides to control fungal infestations and sustain yields (Zubrod et al., 2019). These compounds already account for >35 % of the global pesticide market and their use is predicted to continue rising (Research and Markets, 2015; Zubrod et al., 2019). Fungicides are considered harmless to honey bees due to their low acute toxicity, and are therefore authorized for use during full bloom periods (Almasri et al., 2020). Difenoconazole (DIFENO) is a systemic triazole fungicide that acts as a sterol biosynthesis inhibitor (SBI fungicides), causing morphological and functional changes in the fungal cell membrane (Guo et al., 2010).

Although most studies focus on individual products or active ingredients, pollinators are often exposed to complex mixtures of agrochemicals. These exposures can include chronic low-level exposure to systemic insecticides and acute exposure to fungicides applied at high concentrations (Martins et al., 2023). Furthermore, the toxicity of pesticides to bees may be increased when combined with adjuvants, such as surfactants and oils, which are used to boost pesticide absorption and distribution on plants (Encerrado-Manriquez et al., 2024). In addition, some pesticides, such as the herbicide glyphosate (GLY) and the insecticide chlorantraniliprole (CHLOR), do not repel bees, increasing the likelihood of exposure. Traces of GLY were found in pollen and honey samples, while cyhalothrin residues were found in pollen, wax, brood cells, and bees (Berg et al., 2018; Bommuraj et al., 2019; Macri et al., 2024; Mullin et al., 2010; Rubio et al., 2014;). In turn, CHLOR residues have been found in beebread, honey, and forager bees (Macri et al., 2024; Wang et al., 2022). Meanwhile, traces of the fungicide DIFENO were found in bee bodies, brood, pollen, and honey (Rondeau and Raine, 2022).

From an analytical point of view, apicultural samples are considered difficult matrices that contain large amounts of components including fats or proteins. In the last decade, analytical approaches have been improved in terms of selectivity and sensitivity with wider scopes for the determination of multiple chemical residues in honeybee, beeswax, beebread and honey (Morales et al., 2020). Liquid chromatography (LC) and gas chromatography (GC) coupled to tandem mass spectrometry instruments have been widely used for the determination of pesticide residues in apicultural matrices due to their high specificity and low limits of detection (Bargańska et al., 2013; Morales et al., 2020; Nassar et al., 2024; Sadowska et al., 2019).

Several studies have demonstrated that some pesticides combinations can produce synergistic toxicity effects on bees (Johnson et al., 2013; Pilling et al., 1995; Thompson and Wilkins, 2003; Thompson et al., 2014; Wang et al., 2020). A well-documented interaction occurs between SBI fungicides and different types of insecticides. SBI inhibit cytochrome P450-monooxygenase, a key enzyme in bee metabolic detoxification processes, thereby enhancing insecticide toxicity (Berenbaum and Johnson, 2015; Carneseccchi et al., 2019). As a result, the expression profiles of these pesticide-response biomarker genes can serve as indicators of stress in larvae, even in the absence of symptoms (Vázquez et al., 2020). Lastly, recent meta-analyses suggest that the sublethal effects of fungicides on honey bees have been relatively understudied, especially under field conditions (Cullen et al., 2019; Di Noi et al., 2021).

With these considerations in mind, our study aims to evaluate the impact of sequentially applied pesticide mixtures and the changes in the surrounding flora on honey bee larval detoxification pathways and colony foraging activity under field-realistic conditions. Furthermore, we analyzed the correlation between these factors and the presence of pesticide traces in hive products such as honey and beeswax, as well as the composition of agricultural landscapes surrounding the beehives.

2. Materials and methods

2.1. Study site

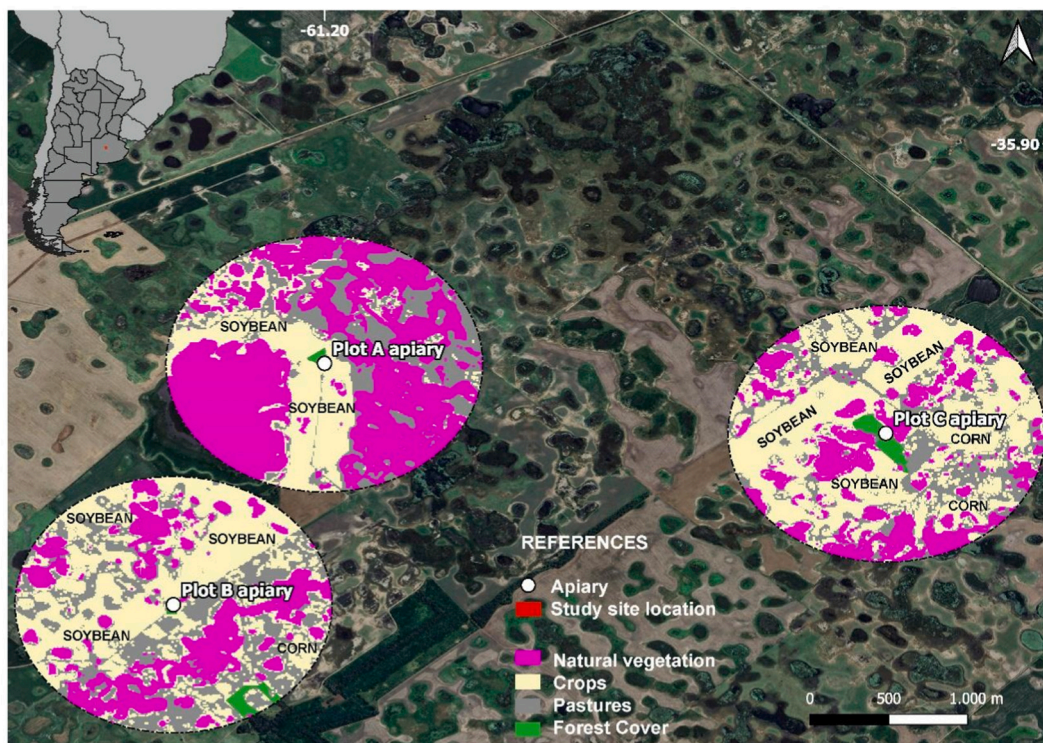
Behavioral assays, and sample collection were conducted during the summer, in hives of European honey bees (*Apis mellifera* L.) located at a farm near Carlos Casares, Buenos Aires Province, in the humid Pampean region of Argentina (35°57'07.29" S, 61°13'06.33" O) (Fig. 1). The study area featured two genetically modified (GM) crops: soybean (Nidera 4611) and corn (KWS KM 4500 GL Stack), which were sown in October 2018. Both crops were herbicide tolerant.

Post-emergence weed control on the different crops was carried out at the beginning of January with a land sprayer (1.5 L/ha), before crop flowering. Subsequently, a mix of fungicide and insecticide was applied simultaneously at the beginning of February, by aerial application (40 cc/ha), and after crop flowering. The first sampling was carried out 3 days before post-emergence weed control and 20 days before aerial application of the fungicide-insecticide mixture. The second sampling was conducted 25 days after herbicide application and 7 days after fungicide-insecticide application. The pesticides used were a GLY-based herbicide (SULFOSATO TOUCHDOWN® SYNGENTA), an insecticide based on Chlorantraniliprole and lambda cyhalothrin (AMPLIGO®), and a difenoconazole-based fungicide (MIRAVIS DUO®) (Table S1).

Weather conditions during both sampling events were optimal for the bee foraging activity, i.e., mean daily temperature of 29 and 30 °C and mean daily RH of 56 and 57 %, respectively. During the first sampling, all crops were in their vegetation stage (Fehr et al., 1971; Ritchie and Hanway, 1982), and the wild flora from uncultivated areas was the only food source for bees. In contrast, during the second sampling, all crops were in full bloom.

Three distinct crop settings (plots) were chosen according to the presence of an apiary in their surroundings. The apiaries were separated by at least 2.5 km. According to previous studies, honey bee colony conducts much of their foraging behavior in agricultural settings within several hundred meters of the hive on average, while in forest areas this behavior shows a modal, median and mean distance of 0.7, 1.6 and 2.2 km around (Seeley, 1995; Vaissière et al., 2010). It is unlikely that foraging areas overlapped between apiaries, though this cannot be

A



B

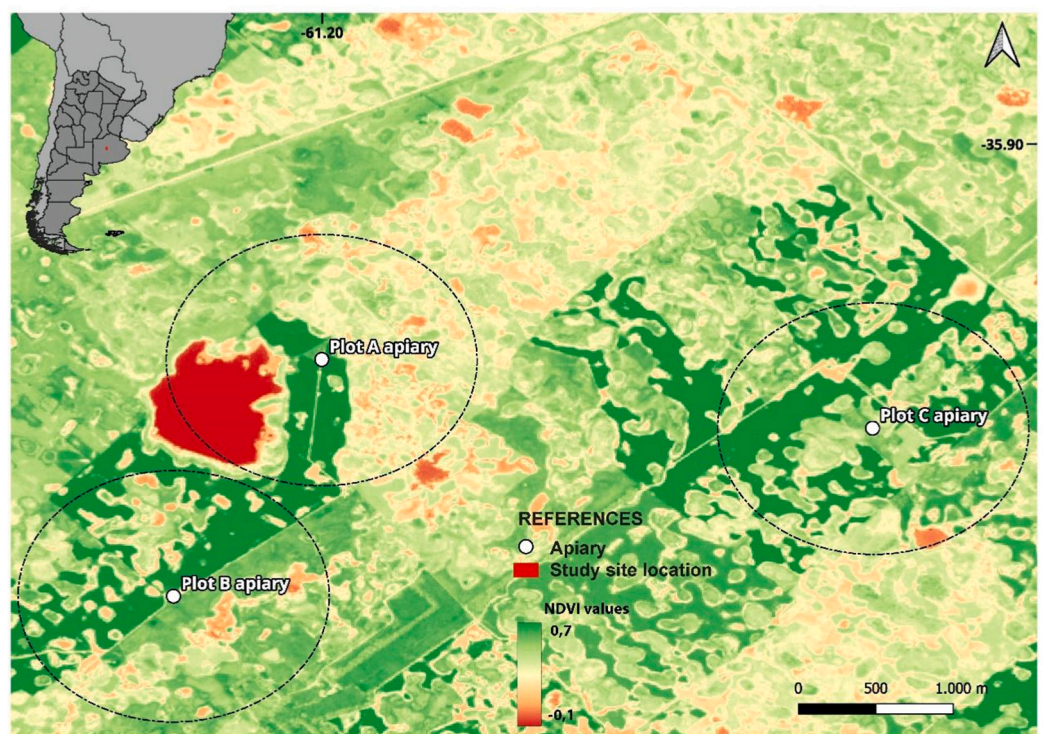


Fig. 1. Geographical location of the three studied apiaries at San Claudio Farm, Carlos Casares, Province of Buenos Aires, Argentina. A satellite image corresponding to the sampling dates was obtained from the European Space Agency catalog (ESA - <https://dataspace.copernicus.eu/>). **A)** The map illustrates the potential foraging areas of three studied apiaries, represented as circles of 1 km radius (316 ha), with different colors indicating the occupied area (ha) for each land cover category, as analyzed using QGIS software. **B)** The map displays the NDVI (normalized differential vegetation index) values, which visually represent the phenological state of the vegetation in the areas surrounding the apiaries.

completely ruled out.

2.2. Spatial characterization of the study area using GIS

The study area was analyzed using QGIS software. To classify the landscape, unsupervised classifications were made from a sentinel satellite image (10 m of spatial resolution) obtained from the European Space Agency catalog (ESA - <https://dataspace.copernicus.eu/>). Moreover, a normalized differential vegetation index (NDVI; Tucker, 1979) was applied to visually interpret the phenological state of the vegetation surrounding the study sites and to provide information on potential foraging resources for bees. Thus, occupied area (ha), proportion, and NDVI (normalized differential vegetation index) for the different cover categories of the three apiaries studied, were obtained in QGIS software for its potential foraging area (1 km of radius). In addition, inverse Simpson and Shannon diversity indices were calculated for the surrounding of each apiary.

2.3. Colony activity

To analyze bee foraging-related behavior under natural conditions, we employed a Before-After (BA) design, a robust approach for evaluating environmental effects on ecological variables (Wauchope et al., 2021). As an indicator of colony activity, we measured the incoming rate, defined as the number of bees entering the hive per unit time at each sampling event (Baker and Jay, 1973). In each apiary, eight colonies with similar population sizes were randomly selected. These colonies consisted of a mated queen, three or four brood frames, and food reserves. The same eight hives were monitored at the two sampling moments. For each selected hive, the number of incoming bees was recorded for 1 min (Delaplane et al., 2013; Farina et al., 2023). To distinguish foraging roles, incoming bees were categorized as pollen foragers (bees with visible pollen loads on their hind legs) or non-pollen foragers (bees without pollen loads, typically nectar foragers). Previous reports showed that most of the non-pollen foragers returning to the colony carried nectar in their honey crops, so counts of non-pollen foragers can provide a reasonable estimate of nectar foraging rates (Diaz et al., 2013; Fewell and Winston, 1996).

2.4. Sample collection

To assess the detoxification response of larvae, we sampled five larvae in the fifth instar from four of the eight randomly selected colonies in each of the apiaries. Sampling was conducted before and after pesticide applications. Fifth instar larvae were pooled into a single cryovial *in situ* and immediately placed into liquid nitrogen for subsequent RNA extraction. We sampled larvae in the fifth instar due to their diet, as this developmental stage primarily consumes resources collected in the external environment (honey and bee bread derived from nectar and pollen) (Medrzycki et al., 2008). Additionally, operculated honey and beeswax samples were collected from the same beehives, during both same sampling moments for pesticide analysis.

2.5. Gene expression analysis

The internal physiological state of larvae was evaluated using a set of molecular biomarkers associated with the detoxifying response in *A. mellifera*. Changes in the transcription of these genes are signs of chemical stress associated to pesticide ingestion through food (Macri et al., 2021; Tomé et al., 2020; Vázquez et al., 2020). Total RNA was extracted from a pool of five fifth instar larvae with 100 µl of TRIzol® according to the supplier's protocol, and the extracted RNA was resuspended in 25 µl of DEPC-treated water. Total RNA was treated with DNase (Promega) before reverse-transcription (RT) reactions. The cDNA was synthesized using 2 µg (RNA was quantified using the NanoDrop 2000 equipment from Thermo Scientific™) of treated RNA

per sample by means of the M-MLV reverse transcriptase system (Promega). Thus, we utilized four biological replicates for apiary, with a pool of five larvae in the fifth instar.

The expression levels were measured by quantitative Polymerase Chain Reaction (qPCR) using three technical replicates for each biological sample. The expression of the P450 cytochromes CYP6AS2, CYP6AS3, CYP6AS4, CYP6BD1 and CYP9Q3, involved in the Phase I of the xenobiotic detoxification pathways in honey bees (Berenbaum and Johnson, 2015) was analyzed (Table S2). The analysis was performed using the 2-ΔΔCt method (Livak and Schmittgen, 2001). For the correct use of this method, primer efficiencies were calculated and validated for both targets and housekeeping genes (Table S2). Melting curve analysis was used to ensure a specific amplification. *Rp49s* and *Rpl8* were used as endogenous controls as they were previously described as reliable reference genes for *A. mellifera* (Collins et al., 2004; Grosso et al., 2018; Lourenço et al., 2008; Scharlaken et al., 2008). Expression levels from the first sampling moment were used as controls to compare changes in the relative expression of the selected targets after pesticide application. In all qPCR experiments, non-template controls were included.

2.6. Pesticide residue analysis

Pesticide residues in honey and beeswax were analyzed, before and after crop flowering and pesticide applications. The direct residue determination of GLY was performed on a 1 g of sample, which was previously homogenized and centrifuged, with the addition of 4.3 ml of extraction solvent, containing 50 mM acetic acid and 10 mM EDTA. In this way, samples were extracted with an aqueous solution, centrifuged, and analyzed without going through tedious and time-consuming derivatization and concentration (Chamkasem et al., 2016). The direct residue determination of herbicide ATZ, fungicide DIFENO and the insecticides Lambda Cyhalothrin (LCT) and CHLOR was performed using the QuEChERS extraction method (quick, easy, cheap, effective, rugged, and safe) (Anastassiades et al., 2007) on a 1 g of sample. This amount of sample was shaken, homogenized, and centrifuged, with the addition of 5 ml of acetonitrile, 3 g of magnesium sulphate (MgSO₄), and 0.75 g of sodium acetate (NaOAc) (Phenomenex). In the second step (clean up), an aliquot of the extract (1 ml) was transferred into a single polypropylene centrifuge tube that contained 150 mg of MgSO₄ and 50 mg of Primary/Secondary Amine (PSA) sorbent, which was shaken and centrifuged for 5 min at 2500 revolutions per minute (rpm). The supernatant was collected in chromatography vials. The extracts obtained were analyzed for pesticide residues with gas chromatography–mass spectrometry (GC–MS) and liquid chromatography–mass spectrometry (LC–MS), according to the type of agrochemical.

Liquid chromatography analyses for GLY determination were performed using a Waters Acquity Ultra Performance Liquid Chromatography (UHPLC) apparatus equipped with a single quadrupole mass detector (waters SQD) and the column HyperCarb Thermo 3 µm 2.1 mm × 100 mm (Thermo-Fisher). The mobile phase used was 0.1 % acetic acid in water: 0.1 % acetic acid in acetonitrile at the following gradient: initial condition during 2 min was (95:5) - (95:5), (95:5)-(0:100) 2–15 min and (95:5) - (95:5) 15–20 min. The selected ion-monitoring mode was used in quantification analysis. The mass-spectrometer acquisition settings were ESI positive and retention time and abundance of the confirmation ion (Ion C) relative to that of the quantification ion (Ion Q) were used as identification criteria. The limit of detection (LOD) was 20 µg/kg, and the limit of quantification (LOQ) was 40 µg/kg (Table S3). The same equipment was used in the case of DIFENO determination, using an Xbridge BEH C18 2.5 µm 2.1 × 150 mm column. The mobile phase used was 0.1 % acetic acid in water:methanol at the following gradient: initial condition during 2 min was (95:5) - (95:5), (95:5) - (80:20) 2–5 min, (80:20) - (20:80) 5–10 min, (20:80) - (0:100) 10–11 min, (0:100) - (0:100) 11–13 min, (0:100) - (95:5) 13–14 min, and (95:5) 14–20 min. As well as for GLY the limit of detection was 20 µg/kg, and the limit of quantification was 40 µg/kg (Table S3).

Gas chromatography analyses for herbicide ATZ and the insecticides CHLOR and LCT, determination were performed using a Perkin Elmer Clarus 600 gas chromatography (GC) apparatus equipped with a single quadrupole mass detector. Sample volumes of 2.0 µl were injected into the programmable split/split less injector, in split less mode with the split outlet opened after 1.5 min with injector port temperature at 250 °C. The capillary column used was DB-5MS (30 m 0.25 µm I.D.; 0.25 mm film thickness) (Agilent Technologies). Helium carrier gas was programmed with a constant flow of 1 ml/min. The oven temperature program was initially set at 80 °C with 5 min hold and ramped (80–178) °C at 25 °C/min, (178–205) °C at 2 °C/min, 205–300 °C at 30 °C/min with a hold of 5 min. The selected ion-monitoring mode was used for quantification analysis. The mass-spectrometer acquisition settings were electron impact ionization 70 eV; retention time and abundance of the confirmation ion relative to that of the quantification ion were used as identification criteria. The limit of detection for ATZ was 5 µg/kg, and the limit of quantification was 10 µg/kg. In the case of the insecticides CHLOR and LCT, the limit of detection was 2 µg/kg, and the limit of quantification was 5 µg/kg (Table S3).

To determine the concentrations of pesticide residues, matrix-matched calibration curves were constructed in the range of 50 to 200 µg/kg for GLY and DIFENO, 20–200 µg/kg for ATZ, and 10 to 30 µg/kg for both insecticides (Table S3). The calibration curves were processed together with the samples to ensure the correct performance of the analysts and the chromatographs. Concentrations were calculated by interpolation. The LOQ was established as the lowest concentration level of the matrix-matched calibration curve, because it meets the reproducibility criterion (European Commission, 2020) and has a signal–noise ratio >10 (Table S3). Selectivity, linearity, accuracy, precision, uncertainty and sensibility were studied in the laboratory for the validation of the methods applying satisfactorily the criteria established in EU-SANTE Guide 2021 (European Commission, 2020). Linearity was evaluated for each compound analyzed as the correlation coefficients (R) obtained from the calibration curves constructed during this study. A recovery study was carried out to evaluate accuracy as recovery (R %) (Table S3).

In all cases, measurements below LOQ were taken as zero concentration for statistical analyses.

2.7. Multivariate and correlation analyses

To explore the underlying relationship between the detoxifying responses in larvae, the presence of pesticide residues in honey and wax samples, and the gathering of food resources after crop flowering and pesticide applications, we performed correlation analyses using Spearman test, suitable for data with non-normally distributed residuals, and Principal Component Analysis (PCA). These analyses were made with the data obtained in the sampling carried out after crop flowering and pesticide applications, focusing on the same four randomly selected colonies from each apiary.

2.8. Statistics

All statistical tests were performed with R v3.5.1 (R Core Team, 2018) using the *glmer* function for mixed models (GLMM) of the “lme4” package (Bates et al., 2009). The “glmmTMB” package (Brooks et al., 2017) was used to determine the Binomial error distribution. Colony activity rates were analyzed with a GLMM following a Poisson error distribution. Gene expression data from qPCR were analyzed by means of a GLMM with gamma error distribution. Non-parametric tests were used in the case of genes that do not meet the assumptions for a gamma distribution of the error. A paired Wilcoxon rank test for comparison between sampling moments was used, while Friedman rank sum test was used for comparison between plots, both with an α level of 0.05. The Conover *post-hoc* test was used for pairwise comparisons between plots and a corrected *p*-value with the Bonferroni procedure. The Wilcoxon

and Friedman tests are recommended for paired data whose errors do not follow a normal distribution (Dalggaard, 2008). The same methods were used in the case of pesticide analysis data. All non-parametric tests were carried out using the “PMCMRplus” package (Pohlert, 2020). The package “vegan” was used to estimate the diversity index (Oksanen et al., 2019).

All GLMs were inspected for over/under dispersion and, scaled residuals were simulated from the fitted model using the DHARMA packages (Hartig, 2020). In all mixed models, the factor “hive” was included as a random variable. The significance of terms within the models was tested using the anova function to compare between nested models, starting from the highest-order terms (Chambers and Hastie, 1992). Non-significant terms ($p > 0.05$) were sequentially removed. Tukey’s *post hoc* test tests were conducted for pairwise comparisons using the “emmeans” package (Lenth, 2020). Assumption of homoscedasticity and normality were checked using Levene and Shapiro–Wilk tests, respectively.

PCA and Spearman’s rank test for correlations between studied variables were performed using the R libraries “ade4” (Thioulouse et al., 2018) and “psych” (Clark, 2020), respectively. Bonferroni correction was applied for multiple comparisons. “FactoMineR” package (Lê et al., 2008) was used to generate PCA graphics.

3. Results

3.1. Spatial characterization of the study area using GIS

Based on the spectral responses of the data, the landscape was categorized into four classes: Crops, Natural vegetation, Pastures and Forest cover (Fig. 1, Table S4). The surrounding of the three apiaries varied in the crop proportion and type of uncultivated areas (Fig. 1; Table S4), most of which are melliferous vegetation (Table S5). In increasing order of crop proportion, Plot A apiary contained 25 hives and was located in a forest patch. This plot covered 189 ha, with crops occupying 23 % of the honey bees navigation area. The main crop was first-planting soybean. Plot B apiary housed 42 hives and was also situated in a forest patch. This plot had an area of 178 ha, with crops occupying 42 % of the navigation area, primarily first-planting soybean. Plot C apiary contained 32 hives and was in a forest patch with a wetland area. This plot had an area of 189 ha, with crops occupying 51 % of the navigation area, including soybeans and corn (Fig. 1, Table S4). In addition, the calculated diversity indices show that the surroundings of the Plot B apiary were more diverse than those of plot A and C (Table S1), although the difference was not significant.

3.2. Colony activity

No significant differences were observed in the total number of bees returning to the hive, either between sampling moments (Fig. 2A) or among plots. (Fig. S1; Table S6). In contrast, the statistical analysis revealed a significant decrease in the number of bees returning to the hive with pollen loads per unit of time during the second sampling moment (Poisson Model: Pollen foragers’ incoming rate ~ Moment + (1|Hive); LRT = 63.929; $p < 0.001$; Fig. 2B). However, no significant differences were detected among plots for this variable (Fig. S1; Table S6).

3.3. Relative gene expression

The expression levels of *CYP6AS2* (Gamma model: Gene expression ~ Moment + (1|Hive), LRT = 31.192, $p < 0.001$) and *CYP6AS4* (Gamma model: Gene expression ~ Moment + (1|Hive), LRT = 6.491, $p = 0.011$) were significantly higher after crop flowering and pesticide applications (Fig. 3). In contrast, no significant differences were found between sampling moments for the relative expression of *CYP6AS3*, *CYP6BD1* and *CYP9Q3* (Fig. 3; Table S7). In turn, no significant differences were found between plots for any of the cytochromes P450s studied (Fig. S2;

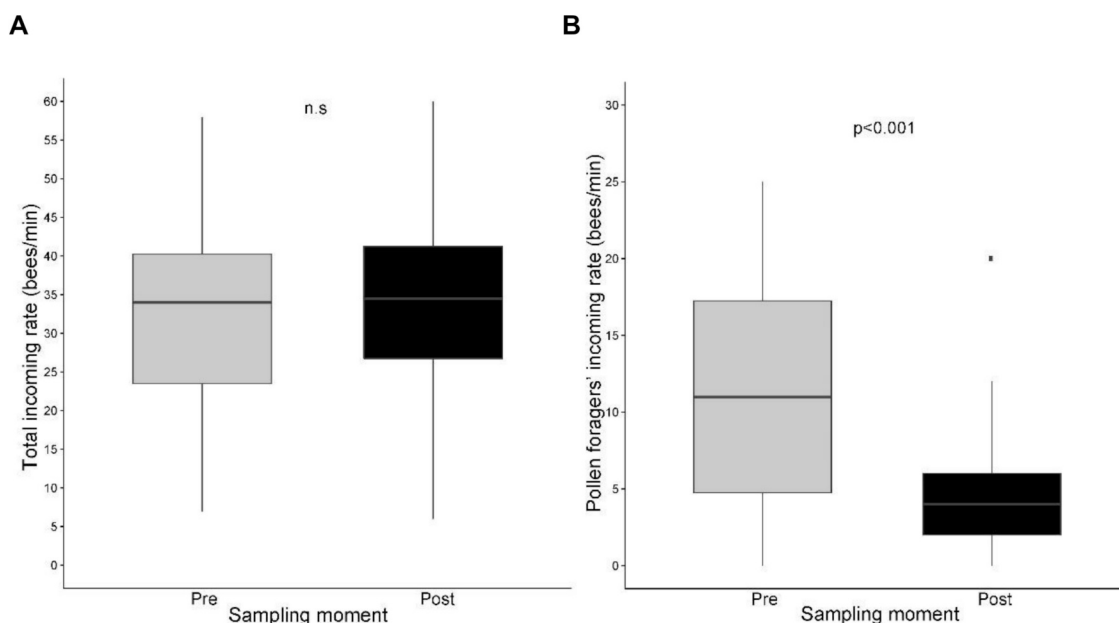


Fig. 2. Colony activity at the hive entrance. A) Total number of bees entering the hive per minute before and after crop flowering and pesticide applications (Poisson Model: Total foragers' incoming rate $\sim 1 + (1|Hive)$; $N = 24$). B) Number of bees entering the hive with pollen loads per minute before and after crop flowering and pesticide applications (Poisson Model: Pollen foragers' incoming rate $\sim \text{Moment} + (1|Hive)$; $N = 24$). Boxplots display the median and interquartile range (IQR), with whiskers extending to the maximum value within $1.5 \times \text{IQR}$, and individual points mark values outside this range. The p -value indicates the levels of significance (n.s.: $p > 0.05$).

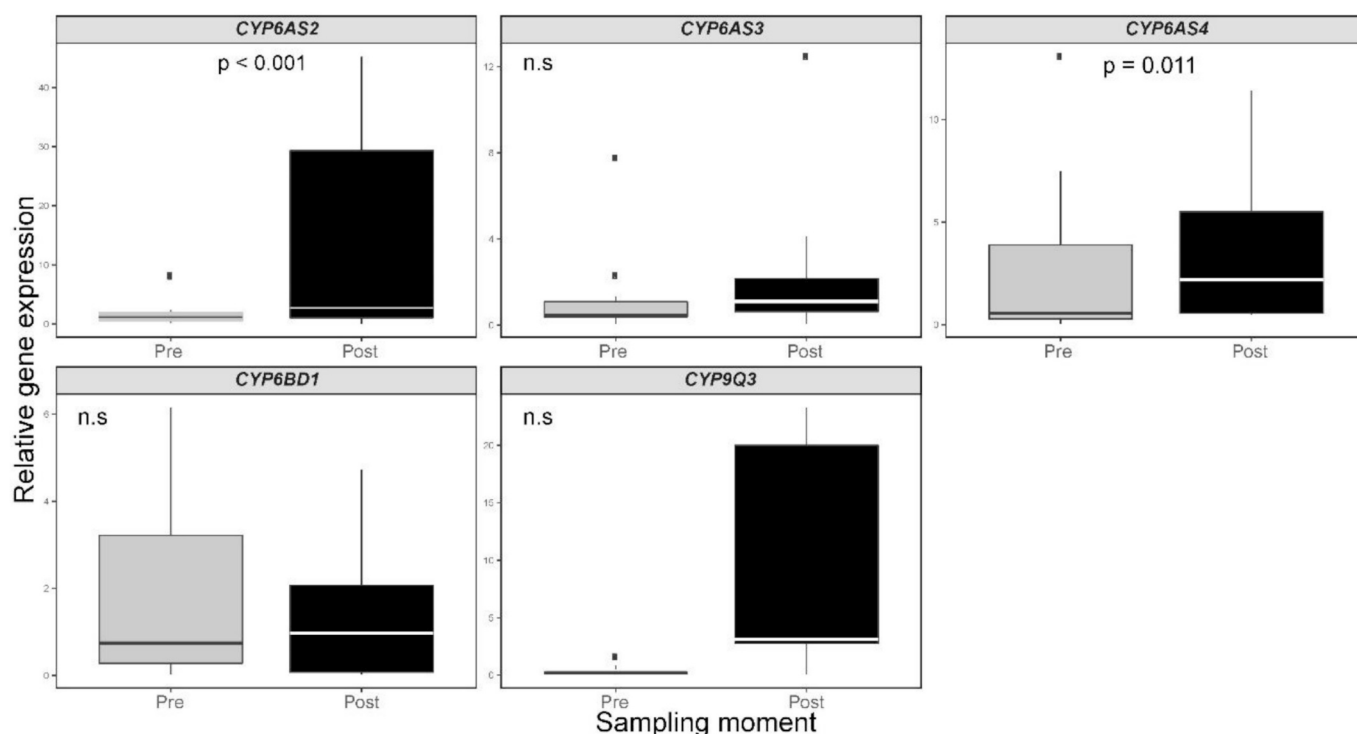


Fig. 3. Relative gene expression of cytochromes P450 in *A. mellifera* larvae before and after crop flowering and pesticide applications. Relative expression levels of the *CYP6AS2*, *CYP6AS3*, *CYP6AS4*, *CYP6BD1* and *CYP9Q3* genes in pools of five fifth instars larvae randomly selected from different colonies. Boxplots display the median and interquartile range (IQR), with whiskers extending to the maximum value within $1.5 \times \text{IQR}$, and individual points mark values outside this range. Gamma model: Gene expression $\sim \text{Moment} + (1|Hive)$; $N = 12$. The p -value indicates significant differences (n.s.: $p > 0.05$).

Table S7).

3.4. Pesticide residue analysis

For honey samples, GLY residues were detected in all samples (Table S8). Although the concentration of GLY was higher after crop

flowering and pesticide application, this difference was not statistically significant (Fig. 4, Wilcoxon test, $V = 26$, $p = 0.308$). Furthermore, no significant differences were observed between plots (Table S9). Regarding insecticides, concentration of CHLOR was significantly higher after crop flowering and pesticide application (Fig. 4, Wilcoxon test, $V = 1$, $p = 0.028$). In contrast, no residues of LCT and the fungicide DIFENO were detected (Table S8).

In the case of beeswax samples, the statistical analysis was limited to the post-flowering and post-application sampling moment due to insufficient sample quantities collected before these events. Thus, a pool of wax samples was analyzed for the studied hives in each apiary. No detectable residues of the herbicide GLY, the insecticide LCT, and the fungicide DIFENO were found in the beeswax samples (Table S10). In contrast, CHLOR residues were detected; however, no significant differences between plots were found after crop flowering and pesticide application (Fig. 5, Table S11).

Lastly, residues of ATZ were detected in all the honey and beeswax samples (Table S8 and Table S10, respectively). In beeswax, significant differences in ATZ concentrations between plots were observed after crop flowering and pesticide application (Fig. 5; Table S11).

3.5. Multivariate and correlation analyses

The Spearman correlation analysis, suitable for data with non-normally distributed residuals, revealed significant associations between pesticide concentrations and the expression of detoxification-related genes. Specifically, CHLOR concentration in honey was negatively correlated with the expression of *CYP6BD1* (Table S12, $Rho = -0.593$; $p = 0.042$). Similarly, CHLOR concentration in beeswax showed a negative correlation with the expression of *CYP6AS2* (Table S12; $Rho = -0.584$; $p = 0.048$). In contrast, the expression level of *CYP6S3* positively and significantly correlated with ATZ concentrations in beeswax (Table S12; $Rho = 0.697$; $p = 0.012$, respectively). Nevertheless, no significant differences were found between colony activity rates and expression levels of detoxification biomarkers in larvae (Table S13).

Concerning PCA, the first three principal components explain 64.94 % of the cumulative proportion of variability (Table S14). The relative expression of *CYP6AS2* and *CYP6AS3* are the variables that better explain the first principal component (PC1), while the ATZ

concentration in honey samples is the variable that better explains the second principal component (PC2). Lastly, GLY and CHLOR concentration in honey samples are the variables that better explain the third principal component (PC3) (Table S15). The dimension, limited by PC1 and PC2, highlighted a clear separation of plot B colonies (represented as red circles), which clustered predominantly in the negative sector of the PC1 (Fig. 6). These two principal components absorbed 27.90 % and 21.50 % of the variability, respectively. The distinction of the plot B colonies from those of plots A (orange circles) and C (pink circles) was associated with higher relative expression of *CYP6AS2* and *CYP6AS3*, and a lower concentration of the insecticide CHLOR in honey and beeswax after crop flowering and pesticide applications.

4. Discussion

The present results show that residues of the pesticides applied in the crop fields located in the surroundings of the studied apiaries enter the colonies and accumulate in hive products such as wax and honey. This study suggests that pesticide exposure may have a significant impact on both the foraging behavior of adult bees and the physiological state of the larvae. The observed reduction in pollen collection, a key indicator of foraging success, likely reflects the broader effects of pesticide exposure on bee behavior, which could lead to reduced colony health and fitness. At the same time, the up-regulation of detoxification-related genes in larvae further supports the idea that these pesticides induce physiological stress, likely because of direct exposure or *via* the food sources they consume. Moreover, the apiary with more diverse surroundings according to the botanical diversity indices obtained exhibit higher relative expression profiles of detoxification-related genes and lower insecticide concentrations in honey and beeswax, after crop flowering and pesticide applications. This may suggest that access to semi-natural floral resources could buffer the effects of pesticide exposure. Taken together, these results highlight the complex interaction between environmental stressors, such as pesticide applications, and the health and functioning of honey bee colonies.

4.1. Impact of pesticide exposure on foraging behavior

Impairment in different variables related to foraging activity of

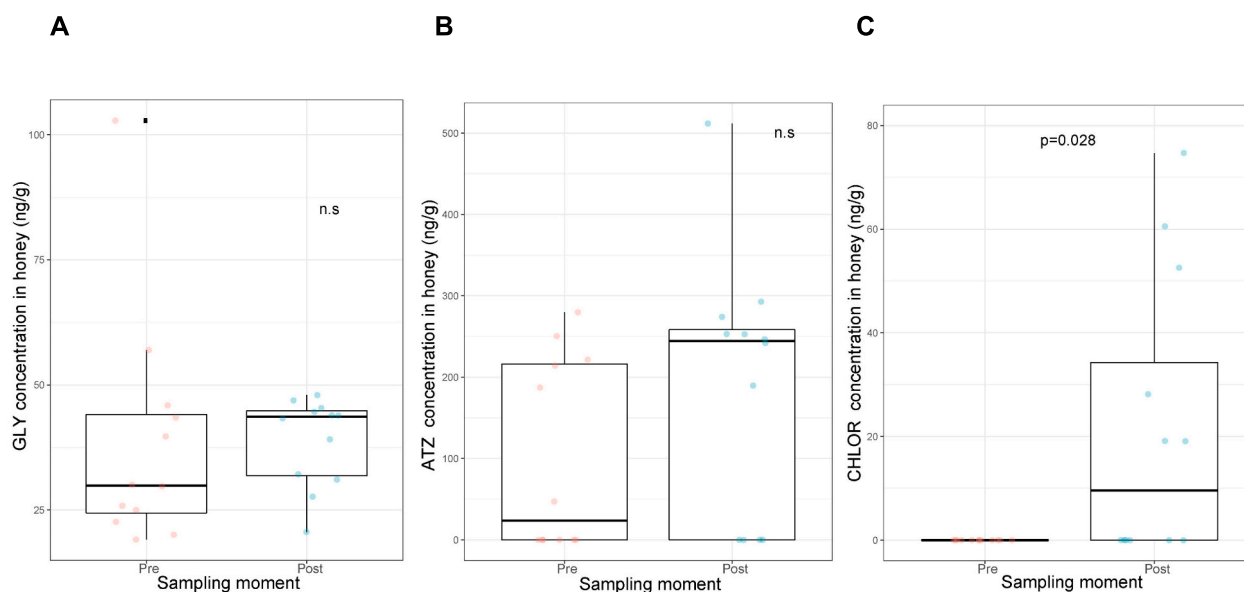


Fig. 4. Pesticide concentration in honey samples before and after crop flowering and pesticide applications. **A)** Glyphosate (GLY) concentration (Wilcoxon test; n.s.; $p > 0.05$; $N = 12$). **B)** Atrazine (ATZ) concentration (Wilcoxon test; n.s.; $p > 0.05$; $N = 12$). **C)** Chlorantraniliprole (CHLOR) concentration (Wilcoxon test; $p = 0.028$; $N = 12$). Boxplots display the median and interquartile range (IQR), with whiskers extending to the maximum value within $1.5 \times \text{IQR}$, and individual points mark values outside this range. The p -value indicates significant differences (n.s.: $p > 0.05$).

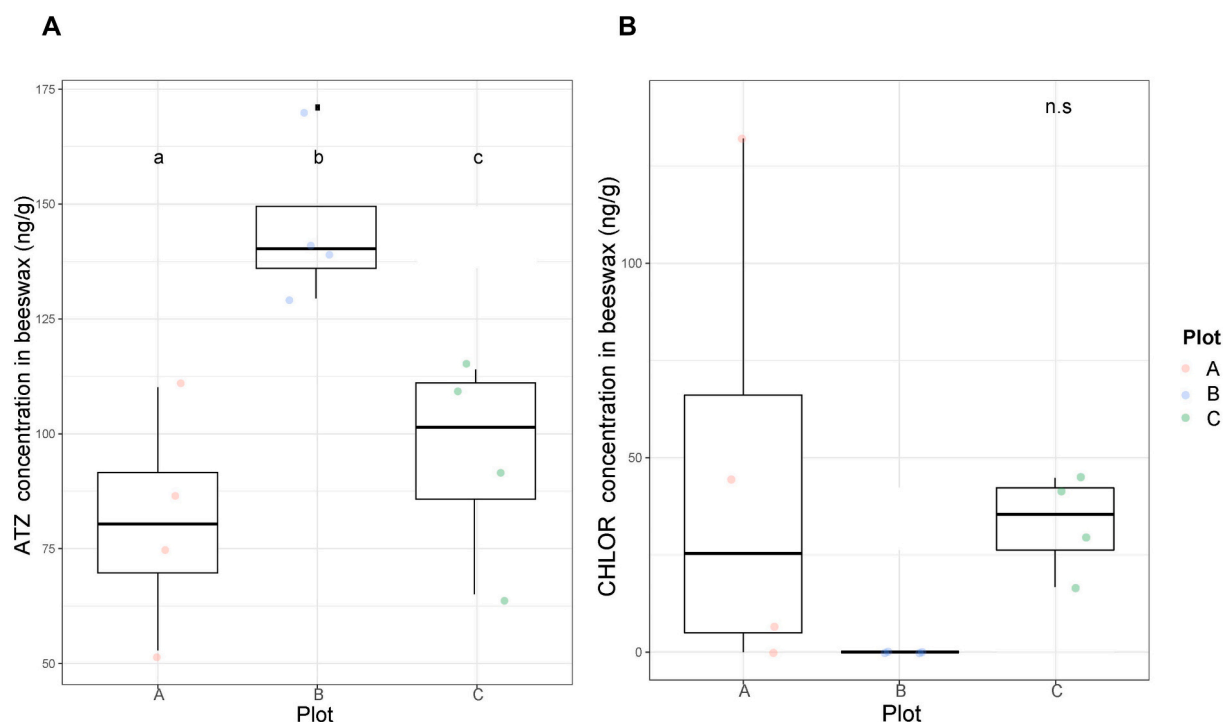


Fig. 5. Pesticide concentration in beeswax samples in each plot and after crop flowering and pesticide applications. A) Atrazine (ATZ) concentration in beeswax (Friedman test; n.s.: $p > 0.018$; $N = 4$). B) Chlorantraniliprole (CHLOR) concentration in beeswax (Friedman test; n.s.: $p > 0.05$; $N = 4$). Boxplots display the median and interquartile range (IQR), with whiskers extending to the maximum value within $1.5 \times \text{IQR}$, and individual points mark values outside this range. Different letters indicate significant differences between plots ($p < 0.05$).

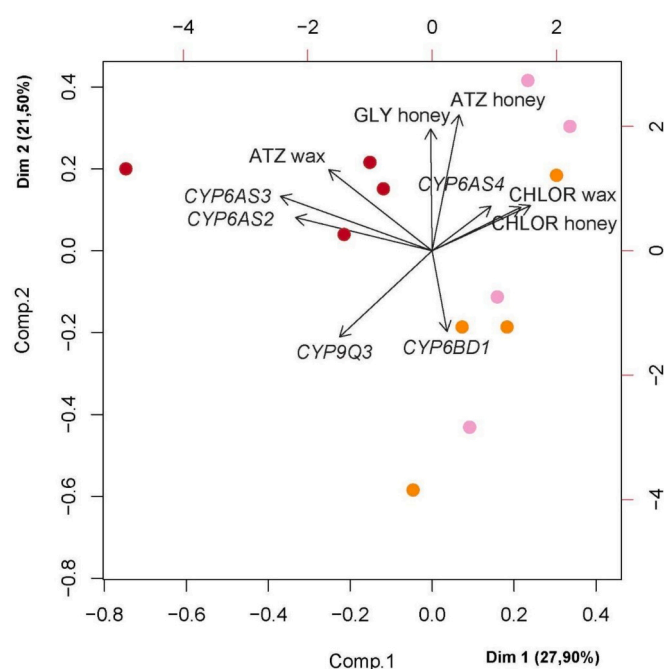


Fig. 6. Principal component analysis (PCA) for relative gene expression and pesticide concentration variables, after crop flowering and agrochemical applications. The individual factor map shows the distribution of apiaries (represented by colored circles) based on the first two principal components (PC1 and PC2), which explain 27.90 % and 21.50 % of the total variability, respectively (see Table S14). Colonies within plot B are represented by red circles, while colonies within plot A and plot C are represented by orange and pink circles, respectively.

honey bees after insecticide exposure have been reported in previous works, such as increasing total foraging trip duration (Schneider et al., 2012; Yang et al., 2008), reducing homing rate (Fischer et al., 2014; Henry et al., 2012; Shi et al., 2019; Tison et al., 2016), decreasing in the number of foraging trips (Colin et al., 2019; Shi et al., 2020; Wu et al., 2023), and -as well as in our study- a reduction of pollen foraging (Wang et al., 2022). The reduction in collection of food resources can affect the brood development and, therefore, impact on the health and productivity of the entire colony (Horn et al., 2021; Picknoll et al., 2021). The impairment of foraging activity thus represents a critical pathway through which pesticide exposure can compromise colony vitality, with cascading effects on colony growth, reproduction, and survival.

4.2. Persistence of pesticide residues and their impact on hive products and non-crop flora

There is evidence that herbicides and their metabolites are highly persistent in the environment, remaining in water and soils for more than a month (Bento et al., 2016; Jablonowski et al., 2011). Consequently, these chemicals are likely to enter the colonies, contaminating bee-derived products and affecting the physiology of both larvae and adult bees (Macri et al., 2021; Macri et al., 2024). We detected residues of agrochemicals applied during crop treatment in both wax and honey samples. Within the colony, after bees collect food from contaminated sources, pesticides may concentrate because nectar is evaporated and condensed by food processor bees to make honey (Seeley, 1995; Rondeau and Raine, 2022). Furthermore, pesticides transferred by contaminated forager bees or pollen loads can also accumulate in beeswax (Nassar et al., 2024), which plays a critical role in the functionality of the hive, serving as material for brood cells and food storage (Michener, 2000). Residues in wax are particularly concerning because larvae develop in direct contact with it for weeks and can also ingest pesticides that migrate from the wax into their diet (Kast and Kilchenmann, 2022). Under controlled rearing conditions, larval survival to adulthood

significantly decreased when developing in GLY-contaminated wax (Tomé et al., 2020). Additionally, beeswax has a low replacement rate, allowing fat-soluble pesticides to accumulate over years, further increasing exposure risks within the colony (Rondeau and Raine, 2022). Remarkably, traces of ATZ, GLY and CHLOR detected in some samples of honey and beeswax exceeded the concentrations considered critical for human consumption, as established by the European Union in honey and other apiculture products (European Commission, 2016). These findings highlight the potential risks posed by pesticide residues in bee products, which may have implications for both environmental health and human safety. Furthermore, residues of ATZ and GLY were also detected in samples collected before the post-emergence pesticide application, suggesting that these herbicides, frequently used in pre-emergence weed control, may have drifted onto the surrounding non-crop flora. Off-target plant contamination can occur through spray drift, runoff, and/or root uptake (Zioga et al., 2022). Such contamination of the wild plant species not only exposes foraging bees to these chemicals but also introduces herbicide residues into the ecosystem, potentially affecting other pollinators and wildlife that depend on these plants for sustenance (Russo et al., 2020; Zioga et al., 2022).

4.3. Molecular impact of pesticide mixtures on larval detoxification pathways

Sublethal effects of ATZ and GLY on honey bee physiology were reported, including the promotion of lipid peroxidation (Jumarie et al., 2017), the impairment of the acetylcholinesterase enzyme activity (Boily et al., 2013), and the carotenoid-retinoid system in honey bees (Helmer et al., 2015). In addition, early symptoms of honey bee intoxication with CHLOR include lethargy, slow and uncoordinated movements and, in some cases, death (Kadala et al., 2019), while short-term exposure to LCT reduced longevity and affected performance of learning and memory in worker bees (Liao et al., 2018). In contrast to these studies focusing on adult bees, our research shifts attention to the molecular effects of pesticide exposure during the larval stage, using a field-based approach. A consistent modulation of immunity and xenobiotic metabolism by GLY ingestion has previously been reported in bee larvae across different *in vitro* experiments (Dai et al., 2018; Gregorc et al., 2012; Tomé et al., 2020; Vázquez et al., 2020). Larvae are more sensitive to the toxic effects of insecticides compared to adult bees (Atkins and Kellum, 1986; Davis, 1989; Heylen et al., 2011; Rortais et al., 2005; Tasei, 2001), which may lead to a gradual weakening of colony structure over time (Zhu et al., 2014). This heightened vulnerability in larvae could reflect their developmental stage, where detoxification mechanisms are less mature, making them more susceptible to chemical stress. By examining the expression of detoxification-related genes in larvae, we provide novel evidence of the impact of pesticide residues in the hive environment on developing individuals.

A synergistic effect between the insecticide CHLOR and triazole fungicides, such as DIFENO, has been reported in acute contact toxicity bioassays, leading to increased cytochrome P450s activity (Haas et al., 2022). While triazole fungicides are well-documented as inhibitors of P450s activity (Berenbaum and Johnson, 2015; Egbuta et al., 2014; Gong and Diao, 2017; Iwasa et al., 2004), they can paradoxically also act as inducers of P450 enzymes under certain conditions (Pal et al., 2022). This dual action suggests that the antagonistic effects of pesticide mixtures, such as those used in this study, may involve both the inhibition and induction of detoxification enzymes. Such induction could enhance the metabolism of pesticide compounds, potentially reducing their toxicity through increased biotransformation (Johnson et al., 2013). Moreover, ATZ exposure has been shown to significantly increase cytochrome P450s activity in honey bees (Fellows et al., 2022), reinforcing the notion that herbicides can play a role in modulating detoxification pathways.

Our results reveal a significant up-regulation of the detoxification-related genes *CYP6AS2* and *CYP6AS4* in larvae after crop flowering

and pesticide applications. This finding contrasts with results from previous studies conducted under field-realistic conditions, where only post-emergence herbicides were applied and no such up-regulation was observed for these genes (Macri et al., 2021). In that study, *CYP6AS4* was downregulated in response to herbicides, while *CYP6AS2* showed no significant changes in expression (Macri et al., 2021). The distinct response observed here suggests that the combined use of multiple pesticides, including insecticides and fungicides, and their sequential application during the crop season may have driven the induction of these genes. The observed induction of *CYP6AS4* aligns with its established sensitivity to pesticide exposure in various experimental contexts, such as larvae reared *in vitro* and those developed within beehives (Macri et al., 2021; Vázquez et al., 2018). However, the up-regulation registered in this study contrast with previous reports of down-regulation, underscoring the potential influence of specific pesticide mixtures. This raises the hypothesis that *CYP6AS4* plays a pivotal role in the detoxification of certain compounds, particularly under conditions of repeated and mixed pesticide exposure. As such, *CYP6AS4* may serve as a dynamic molecular biomarker for larval exposure, reflecting an adaptive response to mitigate the toxic effects of agrochemicals. Future research should focus on the functional characterization of this gene, including its substrate specificity and role in pesticide metabolism.

Altogether, our findings suggest that prolonged exposure to sublethal doses of pesticide mixtures applied throughout the crop season can drive the up-regulation of specific P450-encoding genes in honey bee larvae. The dynamic energy budget theory proposes that environmental stress can significantly affect the energy balance of organisms, leading to increased energy expenditure (Kooijman, 2009). This up-regulation reflects a potential adaptive mechanism to cope with chemical stress within the hive, but it also raises concerns about the metabolic costs and potential trade-offs associated with sustained detoxification activity. A reduction in energy reserves induced by exposure to the insecticide imidacloprid has been reported in honey bee larvae reared *in vitro*, likely associated with the energy consuming activities of *CYP450* detoxification and antioxidant defense (Li et al., 2024). This redistribution of energy resources would lead to an insufficient energy supply for growth and development (Beyers et al., 1999). Investigating these trade-offs and their implications for larval development and overall colony health is essential for understanding the broader impacts of agricultural practices on pollinator populations.

4.4. Multivariate and correlation analyses

Our findings provide additional evidence of pesticide impact by revealing correlations between the concentration of the herbicide ATZ in beeswax and the insecticide CHLOR in both beeswax and honey, and the expression profiles of specific detoxification-related genes in fifth instar larvae. This suggests that the larvae may be activating molecular mechanisms to cope with the stress imposed by pesticide residues present in their hive environment. Moreover, the PCA discriminates the colonies located in the plot B from those of the plot A and C given by higher relative expression profiles of detoxification-related genes and lower insecticide concentrations in honey and beeswax, after crop flowering and pesticide applications. The diversity indices obtained indicate that the surroundings of the plot B apiary were more diverse than those of plots A and C, despite the crop proportions estimated using QGIS. This may reflect the fact that the 58 % non-crop vegetation surrounding plot B was more ecologically diverse compared to the other two apiaries. Such diversity likely provided plot B colonies with access to seminatural floral resources, which may have buffered the effects of pesticide exposure and supported detoxification responses in larvae. However, it is important to note that, despite differences in non-crop vegetation, all colonies were exposed to agricultural environments, underscoring the pervasive risk posed by pesticide use in these landscapes.

Our results highlight the critical role of seminatural vegetation in

agricultural landscapes for mitigating the effects of pesticide exposure on honey bee colonies. SeminatURAL habitats can provide less disturbed foraging areas and help bridge temporary gaps in nectar and pollen availability, as has been previously emphasized (Horn et al., 2021; IPBES, 2016; Ricketts et al., 2008). This study reinforces the need to promote agroecological practices that incorporate habitat diversification and the conservation of seminatural areas to enhance pollinator health and colony resilience (Garibaldi et al., 2014). Moreover, our findings underscore the complex interplay between agricultural practices, pesticide exposure, and detoxification responses in honey bees, particularly during their vulnerable larval stages.

4.5. Limitations

The lack of control over environmental variables, such as local flora, weather conditions, and other external factors, may have influenced the observed results. Additionally, the study focused on the effects of a pesticide mixture, and the potential synergistic and antagonistic interactions among these compounds could have added complexity to the interpretation of their individual effects on bee larvae. Correlations between pesticide exposure and detoxification-related gene expression were observed; however, functional studies are needed to confirm the mechanistic pathways involved for these genes. In addition, the variability in pesticide exposure levels between colonies and within a colony over time could also have contributed to differences in gene expression, potentially complicating the interpretation of these results.

4.6. Conclusions and implications

Most studies on honey bees are carried out under laboratory conditions, with fewer focusing on semi-field or field environments (Di Noi et al., 2021). This study performed under field-realistic conditions highlights the significant impact of pesticide exposure on larval detoxification pathways. Since the survival of honey bee colonies hinges on successful larval development, exposure to chemical residues at critical developmental stages poses a serious risk. Such exposure can lead to irreversible cellular, physiological, and morphological damage, ultimately compromising the transition from larva to healthy adult bees (Morales et al., 2020). Altogether, honeybee colonies permanently exposed to pesticides are likely to show a deficit in information propagation and nectar distribution (Farina et al., 2019). The last recipient in the food distribution chain is the brood, which could also receive food with pesticides in the surroundings studied. A fact that acts as a stressor that affects larval development, which could have implications for overall colony survival in the long term. These findings underscore the need for sustainable agricultural practices that mitigate pesticide exposure to safeguard not only individual colonies, but also the vital ecosystem services that honey bees provide.

CRedit authorship contribution statement

Ivana N. Macri: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Funding acquisition, Formal analysis. **Jose M. Latorre Estivalis:** Writing – review & editing, Writing – original draft, Resources, Methodology. **Denise Nery:** Investigation. **Maria Rosa Derguy:** Investigation. **Diego S. Cristos:** Resources, Methodology. **Jorge A. Zavala:** Writing – review & editing, Conceptualization. **Walter M. Farina:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2025.179388>.

Data availability

Data will be made available on request.

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